但，另一方面：

$$\hat{\rho}_R(1) = \sum_{j=1}^{N_S} \rho_j \, |\widetilde{U_j(1)}\rangle\langle\widetilde{U_j(1)}|$$

$$\Rightarrow \quad \underline{\sum_{\ell=1}^{M} d_{j\ell} d_{j'\ell}^{*} = \rho_j \, \delta_{jj'}}$$

$$\Rightarrow \quad \langle \widetilde{U_j(2)} \,|\, \widetilde{U_i(2)} \rangle$$

$$= \left(\sum_{k'} d_{jk'}^{*} \langle U_{k'}(2)| \right) \left(\sum_{k} d_{ik} \, |U_k(2)\rangle \right) \Big/ \sqrt{\rho_j \rho_i}$$

$$= \sum_{k'} \sum_{k} d_{jk'}^{*} \, d_{ik} \, \langle U_{k'}(2) | U_k(2) \rangle \Big/ \sqrt{\rho_j \rho_i}$$

$\langle U_{k'}(2) | U_k(2) \rangle = \delta_{k'k}$

$$= \sum_{k=1}^{M} d_{jk}^{*} d_{ik} \Big/ \sqrt{\rho_j \rho_i}$$

$$= \rho_j \, \delta_{ij} \Big/ \sqrt{\rho_j \rho_i} = \delta_{ij}$$

证毕.

- $\hat{\rho}_R^{(2)} = ?$ （新表象中）

- $\hat{\rho}(1,2) = |\Psi(1,2)\rangle\langle\Psi(1,2)|$

$$= \left[\sum_{i=1}^{N_s} \sqrt{p_i}\, |\tilde{u}_i(1)\rangle \otimes |\tilde{v}_i(1)\rangle\right]$$

$$\left[\sum_{j=1}^{N_s} \sqrt{p_j}\, \langle\tilde{v}_j(1)| \otimes \langle\tilde{u}_j(1)|\right]$$

- $\hat{\rho}_R^{(2)} = \mathrm{Tr}_1\left[\hat{\rho}(1,2)\right]$

$$= \sum_{l=1}^{N} \langle\tilde{u}_l(1)|\, \hat{\rho}(1,2)\, |\tilde{u}_l(1)\rangle$$

$$= \sum_{i=1}^{N_s} p_i\, |\tilde{v}_i(2)\rangle\langle\tilde{v}_i(2)|$$

类似：$\hat{\rho}_R^{(1)} = \sum_{i=1}^{N_s} p_i\, |\tilde{u}_i(1)\rangle\langle\tilde{u}_i(1)|$

Schmidt 分解

- $$|\psi(1,2)\rangle = \sum_{i=1}^{N_s} \sqrt{p_i}\, |\widetilde{u_i}(1)\rangle \otimes |\widetilde{v_i}(2)\rangle$$

- $$\hat{\rho}_R(1) = \sum_{i=1}^{N_s} p_i\, |\widetilde{u_i}(1)\rangle\langle\widetilde{u_i}(1)|$$

- $$\hat{\rho}_R(2) = \sum_{i=1}^{N_s} p_i\, |\widetilde{v_i}(2)\rangle\langle\widetilde{v_i}(2)|$$

- $\{|\widetilde{u_i}(1)\rangle\}$ 、$\{|\widetilde{v_i}(1)\}$：Schmidt 基

- p_i ：Schmidt 系数 $\left(\sum_{i=1}^{N_s} p_i = 1\right)$

- 非零 p_i 的个数：Schmidt 数.$(\leq N)$

* 当 Schmidt 数为 1 时，$|\psi(1,2)\rangle$ 为可分离态，而两个子系统约化密度算符皆为纯态密度算符.

* 当 Schmidt 数大于 1 时，$|\psi(1,2)\rangle$ 为纠缠态，而两个子系统约化密度算符皆为混态密度算符.

* $\hat{\rho}_R^{(1)}$ 与 $\hat{\rho}_R^{(2)}$ 具有相同的本征谱.

* 纯态 $|\psi(1,2)\rangle$ 的纠缠程度用子系统

Neumann 熵定义：

$$S(\hat{\rho}_R(1)) \equiv -\mathrm{Tr}_1(\hat{\rho}_R(1)\log_2\hat{\rho}_R(1)).$$

在自身表象中，有

$$\hat{\rho}_R(1) = \sum_{i=1}^{N_s} p_i |\tilde{u}_i(1)\rangle\langle\tilde{u}_i(1)| \Rightarrow$$

$$S(\hat{\rho}_R(1)) = -\sum_{i=1}^{N_s} p_i \log_2 p_i$$

(a) 可分离态情况

Schmidt 数为 1，且对应 $p_i = 1$

$\Rightarrow$ $S = 0$

证二

(b) 设子系统维数为 d. 则当

$\hat{\rho}_R(1) = \frac{\mathbb{1}(1)}{d}$ 时, S 取得最大值 $S_{max} = \log_2 d$

· 例: Bell 基 $|\psi_{B1}\rangle$

$$|\psi_{B1}\rangle = \frac{1}{\sqrt{2}}\left[|\uparrow\rangle_1 \otimes |\uparrow\rangle_2 + |\downarrow\rangle_1 \otimes |\downarrow\rangle_2 \right]$$

$$\hat{\rho}_{B1} = \frac{1}{2}\Big[|\uparrow\rangle_1 \otimes |\uparrow\rangle_2 + |\downarrow\rangle_1 \otimes |\downarrow\rangle_2 \Big] \Big[{}_1\langle\uparrow| \otimes {}_2\langle\uparrow| + {}_1\langle\downarrow| \otimes {}_2\langle\downarrow| \Big]$$

$$= \frac{1}{2}\Big[|\uparrow\rangle_1\langle\uparrow| \otimes |\uparrow\rangle_2\langle\uparrow| + |\uparrow\rangle_1\langle\downarrow| \otimes |\uparrow\rangle_2\langle\downarrow| + |\downarrow\rangle_1\langle\uparrow| \otimes |\downarrow\rangle_2\langle\uparrow| + |\downarrow\rangle_1\langle\downarrow| \otimes |\downarrow\rangle_2\langle\downarrow| \Big]$$

$$\Rightarrow \quad \hat{\rho}_{B1}(1)\Big|_R = \frac{1}{2}\left[|\uparrow\rangle_1\langle\uparrow| + |\downarrow\rangle_1\langle\downarrow| \right]$$

$$= \frac{\mathbb{1}(1)}{2}, \quad S = \log_2 2 = 1$$

例： Schmidt 分解 （两个电子自旋系统）

（$\hat{S}_z$-表象； $N=M=2$）

- $|\psi(1,2)\rangle = \frac{1}{\sqrt{2}} \{ \sin\theta\, |\uparrow\rangle_1 \otimes |\uparrow\rangle_2 + \cos\theta\, |\uparrow\rangle_1 \otimes |\downarrow\rangle_2$

$+ \cos\theta\, |\downarrow\rangle_1 \otimes |\uparrow\rangle_2 + \sin\theta\, |\downarrow\rangle_1 \otimes |\downarrow\rangle \}$. （纠缠态）（$\sin\theta \neq \cos\theta$）

即： $C_{11} = \frac{1}{\sqrt{2}} \sin\theta$, $C_{12} = \frac{1}{\sqrt{2}} \cos\theta$

$C_{21} = \frac{1}{\sqrt{2}} \cos\theta$, $C_{22} = \frac{1}{\sqrt{2}} \sin\theta$

$\Rightarrow \hat{\rho}(1,2) = |\psi(1,2)\rangle\langle\psi(1,2)|$

- $\hat{\rho}_R(1) = Tr_2[\hat{\rho}(1,2)] = {}_2\langle\uparrow|\hat{\rho}(1,2)|\uparrow\rangle_2$

$+ {}_2\langle\downarrow|\hat{\rho}(1,2)|\downarrow\rangle_2$

$= \frac{1}{2} [|\uparrow\rangle_1\langle\uparrow| + \sin 2\theta\, |\uparrow\rangle_1\langle\downarrow|$

$+ \sin 2\theta\, |\downarrow\rangle_1\langle\uparrow| + \frac{1}{2} |\downarrow\rangle_1\langle\downarrow|]$

· $\hat{\rho}_R(1)$ 的本征值方程：

$$\hat{\rho}_R(1)\,|\rho\rangle = \rho\,|\rho\rangle$$

· 先在 $\hat{S}_z$ 表象中求解本征值方程 （基矢 $(|\uparrow\rangle, |\downarrow\rangle)$）

$$\begin{cases} \hat{\rho}_R(1) \to \rho_R(1) = \begin{pmatrix} \frac{1}{2} & \frac{\sin 2\theta}{2} \\ \frac{\sin 2\theta}{2} & \frac{1}{2} \end{pmatrix} \begin{cases} \text{密度矩阵} \\ \mathrm{Tr}\,\hat{\rho}_R(1) = 1 \\ \mathrm{Tr}\,\hat{\rho}_R^2(1) < 1 \end{cases} \\ \Rightarrow |\psi(1)\rangle \text{ 为纯态} \\ |\rho\rangle \to \begin{pmatrix} \alpha \\ \beta \end{pmatrix} , \end{cases}$$

本征值方程：

$$\begin{pmatrix} \frac{1}{2} & \frac{\sin 2\theta}{2} \\ \frac{\sin 2\theta}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \rho \begin{pmatrix} \alpha \\ \beta \end{pmatrix} .$$

$$\begin{vmatrix} \frac{1}{2}-\rho & \frac{\sin 2\theta}{2} \\ \frac{\sin 2\theta}{2} & \frac{1}{2}-\rho \end{vmatrix} = 0 \qquad \text{（久期方程）}$$

可得两个本征值和对应本征矢：

$$\begin{cases} \rho_1 = \frac{1}{2}(1 + |\sin 2\theta|) \; ; \; \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \Delta \end{pmatrix} \\ \rho_2 = \frac{1}{2}(1 - |\sin 2\theta|) \; ; \; \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -\Delta \end{pmatrix} \end{cases}.$$

（其中 $\Delta \equiv \frac{\sin 2\theta}{|\sin 2\theta|} = \mathrm{sgn}(\sin 2\theta)$ ）(正交归一)

· 考虑幺正矩阵：取 $U_{11}=\alpha_1$，$U_{12}=\alpha_2$ $U_{21}=\beta_1$，$U_{22}=\beta_2$

$$U \equiv \begin{pmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ \Delta & -\Delta \end{bmatrix} \text{ 及 } U^\dagger \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & \Delta \\ 1 & -\Delta \end{bmatrix},$$

可得，

$$U^{\dagger}\rho_R(1)U=\begin{pmatrix}\rho_1 & 0\\ 0 & \rho_2\end{pmatrix}.$$

· 将本征矢 $\begin{pmatrix}\alpha_1\\ \beta_1\end{pmatrix}$ 所对应的本征右矢用 $|\rho_1\rangle$ 表示，

而本征矢 $\begin{pmatrix}\alpha_2\\ \beta_2\end{pmatrix}$ 所对应的本征右矢用 $|\rho_2\rangle$ 表示.

即
$$\begin{cases}\hat{\rho}_R(1)|\rho_1\rangle=\rho_1|\rho_1\rangle\\ \hat{\rho}_R(1)|\rho_2\rangle=\rho_2|\rho_2\rangle\end{cases}$$

则有，

$$\begin{pmatrix}|\rho_1\rangle\\ |\rho_2\rangle\end{pmatrix}=\underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix}1 & \Delta\\ 1 & -\Delta\end{pmatrix}}_{U^T}\begin{pmatrix}|\alpha_1\\ |\beta_1\end{pmatrix}$$

即
$$\begin{cases} |\rho_1\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_1 + \Delta |\downarrow\rangle_1 \right) \\ |\rho_2\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_1 - \Delta |\downarrow\rangle_1 \right) \end{cases} \quad (\text{正交归一})$$

· 逆变换：

$$\begin{pmatrix} |\uparrow\rangle_1 \\ |\downarrow\rangle_1 \end{pmatrix} = \underbrace{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ \Delta & -\Delta \end{bmatrix}}_{U^*} \begin{pmatrix} |\rho_1\rangle \\ |\rho_2\rangle \end{pmatrix}.$$

即
$$\begin{cases} |\uparrow\rangle_1 = \frac{1}{\sqrt{2}}\left(|\rho_1\rangle + |\rho_2\rangle \right) \\ |\downarrow\rangle_1 = \frac{\Delta}{\sqrt{2}}\left(|\rho_1\rangle - |\rho_2\rangle \right) \end{cases}.$$

· 选 $|\rho_1\rangle, |\rho_2\rangle$ 为基矢 （$\hat{\rho}_R(1)$ 的本征表象）

则有
$$\hat{\rho}_R(1) = \sum_{i=1,2} \rho_i |\rho_i\rangle\langle\rho_i| .$$
（谱表示）

$$|\psi(1,2)\rangle = C_{11}|\uparrow\rangle_1\otimes|\uparrow\rangle_2 + C_{12}|\uparrow\rangle_1\otimes|\downarrow\rangle_2$$

$$+ C_{21}|\downarrow\rangle_1\otimes|\uparrow\rangle_2 + C_{22}|\downarrow\rangle_1\otimes|\downarrow\rangle_2$$

将上式中的 $|\uparrow\rangle$ 和 $|\downarrow\rangle$ 用 $|p_1\rangle$ 和 $|p_2\rangle$ 表示，则有

$$|\psi(1,2)\rangle = C_{11}\left[\frac{1}{\sqrt{2}}(|p_1\rangle + |p_2\rangle)\right]\otimes|\uparrow\rangle_2$$

$$+ C_{12}\left[\frac{1}{\sqrt{2}}(|p_1\rangle + |p_2\rangle)\right]\otimes|\downarrow\rangle_2$$

$$+ C_{21}\left[\frac{\Delta}{\sqrt{2}}(|p_1\rangle - |p_2\rangle)\right]\otimes|\uparrow\rangle_2$$

$$+ C_{22}\left[\frac{\Delta}{\sqrt{2}}(|p_1\rangle - |p_2\rangle)\right]\otimes|\downarrow\rangle_2$$

$$= |p_1\rangle\otimes\left[\left(\frac{C_{11}}{\sqrt{2}} + \frac{C_{21}}{\sqrt{2}}\Delta\right)|\uparrow\rangle_2 + \left(\frac{C_{12}}{\sqrt{2}} + \frac{C_{22}}{\sqrt{2}}\Delta\right)|\downarrow\rangle_2\right]$$

$$+ |p_2\rangle\otimes\left[\left(\frac{C_{11}}{\sqrt{2}} - \frac{C_{21}}{\sqrt{2}}\Delta\right)|\uparrow\rangle_2 + \left(\frac{C_{12}}{\sqrt{2}} - \frac{C_{22}}{\sqrt{2}}\Delta\right)|\downarrow\rangle_2\right].$$

$$= |P_1\rangle \otimes \left[d_{11} |\uparrow\rangle_2 + d_{12} |\downarrow\rangle_2 \right]$$

$$+ |P_2\rangle \otimes \left[d_{21} |\uparrow\rangle_2 + d_{22} |\downarrow\rangle_2 \right].$$

其中，

$$\begin{cases} d_{11} \equiv (C_{11} + C_{21}\Delta)/\sqrt{2} \\ d_{12} \equiv (C_{12} + C_{22}\Delta)/\sqrt{2} \\ d_{21} \equiv (C_{11} - C_{21}\Delta)/\sqrt{2} \\ d_{22} \equiv (C_{12} - C_{22}\Delta)/\sqrt{2} \end{cases}.$$

定义：

$$\begin{cases} |\hat{P}_1\rangle \equiv \left[d_{11} |\uparrow\rangle_2 + d_{12} |\downarrow\rangle_2 \right]/\sqrt{P_1} \\ |\hat{P}_2\rangle \equiv \left[d_{21} |\uparrow\rangle_2 + d_{22} |\downarrow\rangle_2 \right]/\sqrt{P_2} \end{cases}$$

则有，

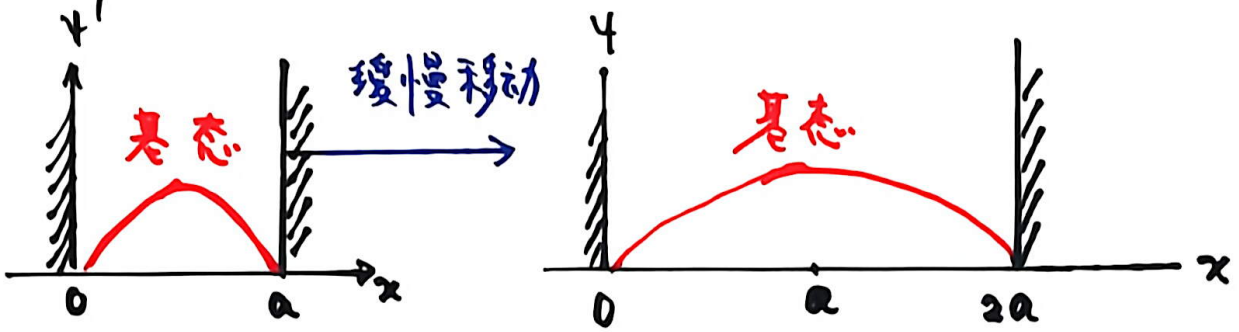
$$\underline{|\psi(1,2)\rangle = \sum_{i=1,2} \sqrt{P_i}\, |P_i\rangle \otimes |\hat{P}_i\rangle}$$

易证：

$$\begin{cases} \langle \hat{P}_1 | \hat{P}_2 \rangle = 0 \\ \langle \hat{P}_i | \hat{P}_i \rangle = 1 \quad (i=1,2) \end{cases}$$

§2.10 绝热近似、Berry相位

· 一维无限深势阱：

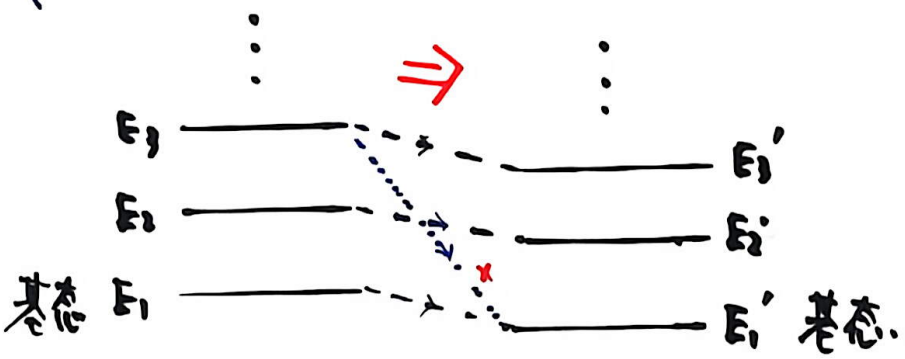


t_0时刻　　t时刻　　$(t>t_0)$

$\hat{H}(t_0)$　　$\hat{H}(t)$

绝热定理

$$\psi_{基态}(x,0) \propto \sin\frac{\pi}{a}x \Rightarrow \psi_{基态}(x,t) \propto \sin\frac{\pi}{2a}x$$



★ 含时薛定谔方程之解的一般形式

· 法1 $|\psi(t)\rangle = \hat{U}(t)\,|\psi(t)\rangle$

· t时刻 $\hat{H}(t)$ 的本征值与本征态 ⇒

· $\hat{H}(t)\,|n,t\rangle = E_n(t)\,|n,t\rangle$ （瞬时）

$$\begin{cases} \langle m,t|\,n,t\rangle = \delta_{mn} \\ \sum_n |n,t\rangle\langle n,t| = \mathbb{1} \end{cases}$$

· $|\psi(t)\rangle = \mathbb{1}\,|\psi(t)\rangle = \sum_n |n\,t\rangle\langle n,t|\,\psi(t)\rangle$

$$\equiv \sum_n C_n(t)\,\exp\left(-\frac{i}{\hbar}\int_0^t dt'\,E_n(t')\right)\,|n,t\rangle$$

$$|\psi(t)\rangle = \sum_n C_n(t)\,e^{i\theta_n(t)}\,|n,t\rangle$$

其中，$\theta_n(t) \equiv -\frac{1}{\hbar}\int_0^t dt'\,E_n(t')$ （动力学相位.）

· $|\psi(t)\rangle = \sum_n C_n(t)\, e^{i\theta_n(t)} |n,t\rangle$.

· $|\psi(t)\rangle$ 代入薛定谔方程中，得 $C_n(t)$ 满足的方程.

· 左边: $i\hbar \frac{d}{dt} |\psi(t)\rangle = i\hbar \frac{d}{dt} \left[\sum_n C_n(t)\, e^{i\theta_n(t)} |n,t\rangle \right]$

$$= \sum_n \left(i\hbar \frac{d}{dt} C_n(t) \right) e^{i\theta_n(t)} |n,t\rangle$$

$$+ \sum_n C_n(t) \left(i\hbar \frac{d}{dt} e^{i\theta_n(t)} \right) |n,t\rangle$$

$$+ \sum_n C_n(t)\, e^{i\theta_n(t)} \left(i\hbar \frac{d}{dt} |n,t\rangle \right) .$$

· 其中 对 $=$ 项的微分为:

$$i\hbar \frac{d}{dt} \left(e^{i\theta_n(t)} \right) = -\hbar\, e^{i\theta_n(t)} \frac{d}{dt} \theta_n(t)$$

$$= -\hbar\, e^{i\theta_n(t)} \frac{d}{dt} \left(-\frac{1}{\hbar} \int_0^t dt' E_n(t') \right)$$

$$= e^{i\theta_n(t)} \epsilon_n(t) .$$

右边：$\hat{H}(t)\left[\sum_n C_n(t) e^{i\theta_n(t)} |n,t\rangle\right]$

$= \sum_n C_n(t) e^{i\theta_n(t)} \hat{H}(t) |n,t\rangle$

$= \sum_n C_n(t) e^{i\theta_n(t)} E_n(t) |n,t\rangle$.

可见，左边未=0又与右边相消. $\Rightarrow$

$$\sum_n e^{i\theta_n(t)} \left[\dot{C}_n(t) |n,t\rangle + C_n(t) \frac{d}{dt} |n,t\rangle \right] = 0 .$$

· 乘以 $\langle m,t|$ $\Rightarrow$

$$\sum_n e^{i\theta_n(t)} \left[\dot{C}_n(t) \underbrace{\langle m,t|n,t\rangle}_{\delta_{m,n}} + C_n(t) \langle m,t| \frac{d}{dt} |n,t\rangle \right] = 0 ,$$

$$\Rightarrow \quad e^{i\theta_m(t)} \dot{C}_m(t) + \sum_n e^{i\theta_n(t)} C_n(t) \langle m,t| \frac{d}{dt} |n,t\rangle = 0 .$$

$$\Rightarrow \quad \dot{C}_m(t) = -\sum_n e^{i\theta_n(t) - i\theta_m(t)} C_n(t) \langle m,t| \frac{d}{dt} |n,t\rangle .$$

- 确定 $C_n(t)$ 的微分方程：

$$\dot{C}_m(t) = -C_m(t)\langle m,t|\frac{d}{dt}|m,t\rangle - \sum_{n\neq m} C_n(t) e^{i\theta_n(t)-i\theta_m(t)}\langle m,t|\frac{d}{dt}|n,t\rangle$$

- 其中 $\langle m,t|\frac{d}{dt}|n,t\rangle$ 可以写成：

$$\langle m,t|\frac{d}{dt}|n,t\rangle = \frac{\langle m,t|\dot{\hat{H}}(t)|n,t\rangle}{E_n(t)-E_m(t)}$$

证明：$\hat{H}(t)|n,t\rangle = E_n(t)|n,t\rangle$.

求以 $\langle m,t|$ $(m\neq n)$ $\Rightarrow$ $\langle m,t|\hat{H}(t)|n,t\rangle = E_n(t)\langle m,t|n,t\rangle = 0$.

上式对时间求导，得

$$\left(\frac{d}{dt}\langle m,t|\right)\hat{H}(t)|n,t\rangle + \langle m,t|\left(\frac{d}{dt}\hat{H}(t)\right)|n,t\rangle + \langle m,t|\hat{H}(t)\left(\frac{d}{dt}|n,t\rangle\right) = 0.$$

$$\Rightarrow \quad E_n(t)\left(\frac{d}{dt}\langle m,t|\right)|n,t\rangle + \langle m,t|\,\dot{\hat{H}}(t)\,|n,t\rangle$$

$$+E_m(t)\langle m,t|\left(\frac{d}{dt}|n,t\rangle\right)=0 \quad .$$

. 但，对 $\langle m,t|n,t\rangle=0$ 求时间导数，得

$$\left(\frac{d}{dt}\langle m|\right)|n,t\rangle + \langle m,t|\left(\frac{d}{dt}|n,t\rangle\right)=0$$

$$\Rightarrow \quad \left(\frac{d}{dt}\langle m,t|\right)|n,t\rangle = -\langle m,t|\left(\frac{d}{dt}|n,t\rangle\right)$$

$$\Rightarrow \quad -E_n(t)\langle m,t|\left(\frac{d}{dt}|n,t\rangle\right) + \langle m,t|\,\dot{\hat{H}}(t)\,|n,t\rangle$$

$$+E_m(t)\langle m,t|\left(\frac{d}{dt}|n,t\rangle\right)=0$$

由此得

$$\langle m,t|\,\frac{d}{dt}\,|n,t\rangle = \frac{\langle m,t|\,\dot{\hat{H}}(t)\,|n,t\rangle}{E_n(t)-E_m(t)} \quad .$$

$(m\neq n)$. 证毕.

$\Rightarrow$

$$\dot{C}_m(t) = -C_m(t)\langle m\,t|\frac{d}{dt}|m\,t\rangle$$

$$-\sum_{n\neq m} C_n(t)\, e^{i\theta_n(t)-i\theta_m(t)} \frac{\langle m\,t|\dot{\hat{H}}(t)|n\,t\rangle}{E_n(t)-E_m(t)},$$

(乎格关系式)

- 由于 $\hat{H}$ 与时间有关，随着时间增加，$|n\neq m\rangle$ 的本征态与 $|m\rangle$ 态会发生混合。

- 设 $\hat{H}(t)$ 随时间的变化足够缓慢，以致上式右方第二项可忽略（绝热近似），则有

$$\dot{C}_m(t) \approx -C_m(t)\langle m\,t|\frac{d}{dt}|m\,t\rangle$$

其解为：$$C_m(t) = C_m(0)\, e^{i\gamma_m(t)}$$

其中 $$\gamma_m(t) \equiv i\int_0^t \langle m\,t'|\frac{d}{dt'}|m\,t'\rangle dt'$$

$\gamma_m(t)$：绝热相位

* 绝热相位 $\gamma_n(t)$ 是实量

证明: $\gamma_n(t) \equiv i\int_0^t dt' \langle n,t'|\dot{n},t'\rangle$

· 证明 $\langle n,t|\dot{n},t\rangle$ 是纯虚的.

由归一化条件: $\langle n,t|n,t\rangle = 1$,

$$\Rightarrow \quad (\dot{\langle} n,t|)|n,t\rangle + \langle n,t|(\dot{|} n,t\rangle) = 0$$

$$\Rightarrow \quad \left[\langle n,t|(\dot{|} n,t\rangle)\right]^* + \langle n,t|(\dot{|} n,t\rangle) = 0$$

$$\Rightarrow \quad 2\mathrm{Re}\left[\langle n,t|\dot{|} n,t\rangle\right] = 0$$

$$\Rightarrow \quad \langle n t|\dot{|} n t\rangle \text{ 为纯虚量!}$$

* 绝热定理 $\Rightarrow$

$\Rightarrow$ 在绝热近似下，有

$$|\psi(t)\rangle = \sum_m C_m(0)\, e^{i\gamma_m(t)}\, e^{i\theta_m(t)}\, |m,t\rangle$$

初始条件：$|\psi(t=0)\rangle = |n,0\rangle$，（$\hat{H}(0)|n,0\rangle = E_n(0)|n,0\rangle$）

$\Rightarrow$ $C_m(0) = \delta_{mn}$.

$\Rightarrow$

$$|\psi(t)\rangle = e^{i\gamma_n(t)}\, e^{i\theta_n(t)}\, |n,t\rangle$$

（$\hat{H}(t)|n,t\rangle = E_n(t)|n,t\rangle$）

绝热定理

✱ 绝热定理成立的条件

- $\dot{C}_m(t) = -C_m(t)\langle m,t|\dot{|m,t\rangle} - \sum_{n\neq m} C_n(t)\, e^{i\theta_n(t)-i\theta_m(t)} \dfrac{\langle m,t|\dot{\hat{H}}|n,t\rangle}{E_n(t)-E_m(t)}$

- 等式右方随时间的变化非常缓慢.

首先，$C_n(t) \approx C_n(t=0) = C_n$，$E_n(t) \approx E_n(t=0) = E_n$

$$\Rightarrow \dot{C}_m(t) = -C_m\langle m,t|\frac{d}{dt}|m,t\rangle - \sum_{n\neq m} C_n\, e^{-\frac{i}{\hbar}(E_n-E_m)t} \frac{\langle m,t|\dot{\hat{H}}(t)|n,t\rangle}{E_n-E_m}$$

- 设 $t=0$ 时，系统处于 $\hat{H}(t=0)$ 的第 k 个本征态 $|k,t=0\rangle$ 上，即 $C_n = \delta_{nk}$，$(k\neq m)$

$$\Rightarrow \dot{C}_m(t) = -e^{-\frac{i}{\hbar}(E_k-E_m)t} \frac{\langle m,t|\dot{\hat{H}}(t)|k,t\rangle}{E_k-E_m}$$

两边对 t 积分 $\Rightarrow$

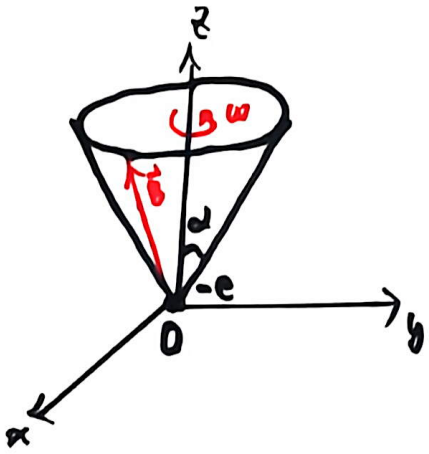
$$\Rightarrow \quad C_m(t) = -\int_0^t d\tau \; e^{-\frac{i}{\hbar}(E_k - E_m)\tau} \frac{\langle m\tau | \dot{\hat{H}}(\tau) | k\tau \rangle}{E_k - E_m}$$

$$\approx \quad \frac{\langle m t | \dot{\hat{H}}(t) | k t \rangle}{E_k - E_m} \frac{i\hbar}{E_k - E_m} \left(1 - e^{-\frac{i}{\hbar}(E_k - E_m)t} \right)$$

$\Rightarrow$ 绝热近似成立的条件，即为 $|C_m(t)| \ll 1$ ，$\Rightarrow$

$$\left| \hbar \frac{\langle m,t | \dot{\hat{H}}(t) | k,t \rangle}{(E_k - E_m)^2} \right| \ll 1$$

绝热近似的简单例子.



例：在磁场强度为 B_0 的磁场中，假设一个电子（电荷为 $-e$，质量为 m）静止在原点，磁场方向与 z 轴夹角为 α，并以常角速度 ω 绕 z 轴转动。设 $t=0$ 时刻粒子处于 $\hat{H}(0)$ 的一个本征态 $\psi(0)=\begin{pmatrix}\cos\alpha/2\\ \sin\alpha/2\end{pmatrix}$，求 t 时刻薛定谔方程的解 $\psi(t)$.

$\hat{S}_B(\sin\alpha, 0, \cos\alpha)$ 方向本征值为 $\frac{\hbar}{2}$ 的本征态

$\left(\hat{S}=\frac{\hbar}{2}\hat{\sigma}\right)$

$$\hat{H}(t)=\frac{e}{m}\vec{B}(t)\cdot\hat{S}=\frac{e\hbar}{2m}\vec{B}(t)\cdot\hat{\sigma}$$

$$\vec{B}(t)=B_0\left[\sin\alpha\cos\omega t\,\vec{i}+\sin\alpha\sin\omega t\,\vec{j}+\cos\alpha\,\vec{k}\right]$$

Pauli表象下，$\hat{\sigma}_x=\begin{pmatrix}0&1\\1&0\end{pmatrix}$, $\hat{\sigma}_y=\begin{pmatrix}0&-i\\i&0\end{pmatrix}$, $\hat{\sigma}_z=\begin{pmatrix}1&0\\0&-1\end{pmatrix}$

$$\hat{H}(t)=\frac{e\hbar B_0}{2m}\left[\sin\alpha\cos\omega t\,\hat{\sigma}_x+\sin\alpha\sin\omega t\,\hat{\sigma}_y+\cos\alpha\,\hat{\sigma}_z\right]$$

$$=\frac{\hbar\omega_0}{2}\begin{pmatrix}\cos\alpha & e^{i\omega t}\sin\alpha\\ e^{i\omega t}\sin\alpha & -\cos\alpha\end{pmatrix},\quad \left(\omega_0\equiv\frac{eB_0}{m}\right)$$

特征频率数字.

· 哈密顿算符的瞬时本征值方程为：

$$\hat{H}(t)\,\chi_n(t)=E_n(t)\,\chi_n(t),$$

易解得两个本征向量和本征值：

$$\begin{cases}\chi_+(t)=\begin{pmatrix}\cos\alpha/2\\ e^{i\omega t}\sin\alpha/2\end{pmatrix}, & E_+=\frac{\hbar\omega_0}{2}\\ \chi_-(t)=\begin{pmatrix}e^{-i\omega t}\sin\alpha/2\\ -\cos\alpha/2\end{pmatrix}, & E_-=-\frac{\hbar\omega_0}{2}\end{cases}$$

$\hbar\omega_0$

(正交归一)

* $E_\pm$ 与时间无关！

⇒ 动力学相位：

$$\begin{cases} e^{i\theta_+(t)} \equiv e^{-\frac{i}{\hbar}\int_0^t dt' E_+(t')} = e^{\frac{-i\omega_0 t}{2}} \\ e^{i\theta_-(t)} \equiv e^{-\frac{i}{\hbar}\int_0^t dt' E_-(t')} = e^{\frac{i\omega_0 t}{2}} \end{cases}$$

- $\psi(t) = C_+(t)\, e^{i\theta_+(t)} \chi_+(t) + C_-(t)\, e^{i\theta_-(t)} \chi_-(t)$

$$= C_+(t)\, e^{-i\frac{\omega_0 t}{2}} \begin{pmatrix} \cos\theta/2 \\ e^{i\omega t}\sin\theta/2 \end{pmatrix}$$

$$+ C_-(t)\, e^{i\frac{\omega_0 t}{2}} \begin{pmatrix} e^{i\omega t}\sin\theta/2 \\ -\cos\theta/2 \end{pmatrix}.$$

将 $\psi(t)$ 代入薛定谔方程可得 C_+, C_- 满足的微分方程. 可以常用方法联立求解满足初始条件的 $C_+(t), C_-(t)$.

可得：

$$\begin{cases} C_+(t) = \left[\cos\left(\frac{\lambda}{2}t\right) - i\frac{(\omega_0 - \omega\cos\alpha)}{\lambda} \sin\left(\frac{\lambda}{2}t\right) \right] e^{-i(\omega-\omega_0)t/2} \\ C_-(t) = i\left[\frac{\omega}{\lambda}\sin\alpha \sin\left(\frac{\lambda}{2}t\right) \right] e^{i(\omega-\omega_0)t/2} \end{cases}.$$

其中 $\lambda \equiv \sqrt{\omega^2 + \omega_0^2 - 2\omega\omega_0\cos\alpha}$.

（多格解）

$\Rightarrow$ 绝热近似、绝热 相位

* 绝热近似的数学处理：

$$\left|\frac{\langle \chi_+(t)|\,\dot{\hat{H}}(t)\,|\chi_-(t)\rangle}{(E_+ - E_-)^2/\hbar}\right| \ll 1$$

$$\Rightarrow \quad \frac{\sin\alpha}{2}\left(\frac{\omega}{\omega_0}\right) \ll 1$$

$$\Rightarrow \quad \boxed{\frac{\omega}{\omega_0} \ll 1}$$

磁场转动的角频率远小于系统特征玻尔频率！

- 当 $\frac{\omega}{\omega_0} \ll 1$ 时，由 $\lambda = \sqrt{\omega^2 + \omega_0^2 - 2\omega\omega_0\cos\alpha}$

得. $\boxed{\lambda \approx \omega_0 - \omega\cos\alpha}$

$\Rightarrow$

- $C_+(t) = \left[\cos\left(\frac{\Lambda}{2}t\right) - i\,\frac{(\omega_0 - \omega\cos\alpha)}{\Lambda} \sin\left(\frac{\Lambda}{2}t\right) \right] e^{-i(\omega-\omega_0)t/2}$

$$\approx \left[\cos\left(\frac{\Lambda}{2}t\right) - i\sin\left(\frac{\Lambda}{2}t\right) \right] e^{-i(\omega-\omega_0)t/2}$$

$$= e^{-i\frac{\Lambda}{2}t}\, e^{-i(\omega-\omega_0)t/2}$$

$$= e^{i(\cos\alpha - 1)\omega t/2} \,,$$

- $C_-(t) = i\left[\frac{\omega}{\Lambda} \sin\alpha \sin\left(\frac{\Lambda}{2}t\right) \right] e^{i(\omega-\omega_0)t/2}$

$$\approx i\left[\frac{\omega}{\omega_0} \sin\alpha \sin\left(\frac{\Lambda}{2}t\right) \right] e^{i(\omega-\omega_0)t/2}$$

- 绝热近似：

$C_-(t) \propto \frac{\omega}{\omega_0} \ll 1$，取 $C_-(t) = 0$

$$\Rightarrow \psi(t) \approx e^{-i\omega_0 t/2}\, e^{i(\cos\alpha - 1)\omega t/2}\, \chi_+(t).$$

满足初始条件：$\psi(0) = \begin{pmatrix} \cos\alpha/2 \\ \sin\alpha/2 \end{pmatrix} = \chi_+(0)$，

$$\begin{cases} \text{绝热相位：} \gamma_+(t) = i\int_0^t dt'\, \chi_+^\dagger(t') \dot{\chi}_+(t') \\ \qquad\qquad = (\cos\alpha - 1)\omega t/2 \\ \text{动力学相位：} \theta_+(t) = -\frac{1}{\hbar}\int_0^t dt' E_+(t') \\ \qquad\qquad = -\frac{1}{\hbar} E_+ t = -\frac{\omega_0 t}{2} \end{cases}$$

$$\Rightarrow \psi(t) = e^{i\theta_+(t)}\, e^{i\gamma_+(t)}\, \chi_+(t)$$

满足绝热定理

存在的问题.

- 在绝热近似下. 有

$$|\psi(t)\rangle = e^{i\gamma_n(t)} e^{i\beta_n(t)} |n,t\rangle.$$

初始条件：$|\psi(t=0)\rangle = |n,t=0\rangle.$

- 绝热相位 $\gamma_n(t)$ 有物理意义吗？

$$\gamma_n(t) \equiv i\int_0^t \langle n,t'|\frac{\partial}{\partial t'}|n,t'\rangle\, dt'$$

- 当改换基矢

$$|n,t\rangle \longrightarrow e^{i\eta_n(t)}|n,t\rangle = |n,t\rangle'$$

时，$\gamma_n(t)$ 怎么变化. （$\eta_n(t)$：任意相位）

$$\hat{H}(t)|n,t\rangle' = E_n(t)|n,t\rangle'$$

- $\{|n,t\rangle'\}$：正交. 完备.

- $|\psi(t)\rangle = \sum_n c_n(t)\, e^{i\theta_n(t)} |n, t\rangle'$

- 代入薛定谔方程，得系数 $c_n(t)$ 满足的方程：

$$\dot{c}_m(t) = -c_m(t) \langle m, t| \frac{\partial}{\partial t} |m, t\rangle' .$$

（绝热近似）

计算 $\langle m t| \frac{d}{dt} |m t\rangle'$ ：

- $\frac{d}{dt}|m t\rangle' = \frac{d}{dt}\left(e^{i\eta_m(t)} |m, t\rangle \right)$

$$= i\left(\frac{d}{dt}\eta_m(t)\right)|m, t\rangle' + e^{i\eta_m(t)} \frac{d}{dt}|m t\rangle ,$$

$$\Rightarrow \quad \langle m t| \frac{\partial}{\partial t} |m t\rangle' = i\frac{d}{dt}\eta_m(t) + \langle m, t| \frac{d}{dt} |m, t\rangle .$$

$$\Rightarrow \quad \dot{c}_m(t) = -c_m(t)\left[i\frac{d}{dt}\eta_m(t) + \langle m, t| \frac{d}{dt} |m t\rangle \right]$$

$$\Rightarrow \quad c_m(t) = c_m(0)\, e^{i\gamma_m'(t)}$$

其中 $\gamma_m'(t) = \gamma_m(t) - \left(\eta_m(t) - \eta_m(0)\right)$

初始值： $c_m(0) = \delta_{mn}$ $\Rightarrow$

$$|\psi(t)\rangle = e^{i\gamma_n'(t)}\, e^{i\theta_n(t)}\, |n;t\rangle'.$$

- $\gamma_n(t)$ 不具备规范不变性 （不可观测的）.
- $\gamma_n'(t) \equiv \gamma_n(t) - [\eta_n(t) - \eta_n(0)]$
- 可取 $\eta_n(t) - \eta_n(0) = \gamma_n(t)$, 则

$$\gamma_n'(t) = 0.$$

$$\Rightarrow \quad |\psi(t)\rangle = e^{i\theta_n(t)}\, |n;t\rangle'$$

（1984年之前）

Berry 的工作 (1984)

英国 Bristol 大学

- $\hat{H}(t) = \hat{H}(\vec{R}(t))$ ← 参数

- $\hat{H}(\vec{R}(t))\,|n, \vec{R}(t)\rangle = E_n(\vec{R}(t))\,|n, \vec{R}(t)\rangle$

- $\langle m, \vec{R}(t) | n, \vec{R}(t)\rangle = \delta_{nm}$

$$\sum_n |n, \vec{R}(t)\rangle\langle n, \vec{R}(t)| = \mathbb{1}$$

- $|\psi(t)\rangle = \sum_n C_n(t)\, e^{i\theta_n(\vec{R}(t))}\, |n, \vec{R}(t)\rangle$

$$\left(\theta_n(\vec{R}(t)) = -\frac{1}{\hbar}\int_0^t E_n(\vec{R}(t'))\,dt'\right)$$

动力学相位

- 周期：T

$$\begin{cases} \vec{R}(t+T) = \vec{R}(t) \\ \hat{H}(\vec{R}(t+T)) = \hat{H}(\vec{R}(t)) \\ |n, \vec{R}(t+T)\rangle = |n, \vec{R}(t)\rangle \end{cases}$$

- 初始条件： $|\psi(0)\rangle = |n, \vec{R}(0)\rangle$

绝热近似下，有

$$|\psi(t)\rangle = e^{i\theta_n(\vec{R}(t))}\, e^{i\gamma_n(\vec{R}(t))}\, |n, \vec{R}(t)\rangle.$$

其中，$\gamma_n(\vec{R}(t)) \equiv i\int_0^t dt' \langle n\vec{R}(t')| \frac{d}{dt'} |n\vec{R}(t')\rangle$.

（绝热相位）

- $|n, \vec{R}(t)\rangle' \equiv e^{i\eta_n(t)} |n, \vec{R}(t)\rangle$

✱ $|n, \vec{R}(T)\rangle' = |n, \vec{R}(0)\rangle'$

$$\Rightarrow \quad e^{i\eta_n(T)} |n, \vec{R}(T)\rangle = e^{i\eta_n(0)} |n, \vec{R}(0)\rangle$$

$$= e^{i\eta_n(0)} |n, \vec{R}(T)\rangle$$

$$\Rightarrow \quad e^{i\eta_n(T)} = e^{i\eta_n(0)}$$

$$\Rightarrow \quad \eta_n(T) = \eta_n(0) + 2n'\pi \qquad n': \text{整数}$$

$\Rightarrow$ $\eta_n(T) - \eta_n(0) = 2n'\pi$, n'：任意整数.

$\Rightarrow$ $|\psi(t)\rangle = e^{i\gamma_n'(t)} e^{i\theta_n t} |n t\rangle'$.

绝热近似

初始条件 $|\psi(0)\rangle = |n,0\rangle'$

$$\gamma_n'(t) \equiv \gamma_n(t) - (\eta_n(t) - \eta_n(0))$$

· 当 $t=T$ 时

$$|\psi(T)\rangle = e^{i\gamma_n'(T)} e^{i\theta_n T} |n T\rangle'$$

$$= e^{i(\gamma_n(T) - (\eta_n(T) - \eta_n(0)))} e^{i\theta_n T} |n T\rangle'$$

$$= e^{i(\gamma_n(T) - 2\pi n)} e^{i\theta_n T} |n T\rangle'$$

$$= e^{i\gamma_n(T)} e^{i\theta_n T} |n T\rangle'$$

$\gamma_n(T)$ 消不掉！ (一般 $\gamma_n(T) \neq 2\pi n$)

$\gamma_n(T)$：Berry 相位

✱ Berry 相位

$$\gamma_n(\tilde{R}_n(t)) = i\int_0^t dt' \langle n, \tilde{R}(t') \left| \frac{d}{dt'} \right| n, \tilde{R}(t') \rangle$$

(绝热相位)

但，$\langle n, \tilde{R}(t) \left| \frac{d}{dt} \right| n, \tilde{R}(t) \rangle$

$$= \langle n, \tilde{R}(t) | \nabla_R | n, \tilde{R}(t) \rangle \frac{\partial \tilde{R}}{\partial t} ,$$

$$\Rightarrow \quad \gamma_n(\tilde{R}(t)) = i\int_{\tilde{R}(0)}^{\tilde{R}(t)} \langle n, \tilde{R} | \nabla_R | n, \tilde{R} \rangle \cdot d\tilde{R} ,$$

· 设过一周期之后，在参数空间中 $\tilde{R}$ 画出一个闭合曲线 C：

$$\Rightarrow \quad \gamma_n(T) = i\int_{\tilde{R}(0)}^{\tilde{R}(T)} \langle n, \tilde{R} | \nabla_R | n, \tilde{R} \rangle \cdot d\tilde{R}$$

$$= i\oint_C \langle n, \tilde{R} | \nabla_R | n, \tilde{R} \rangle \cdot d\tilde{R} \equiv \gamma_n(C)$$

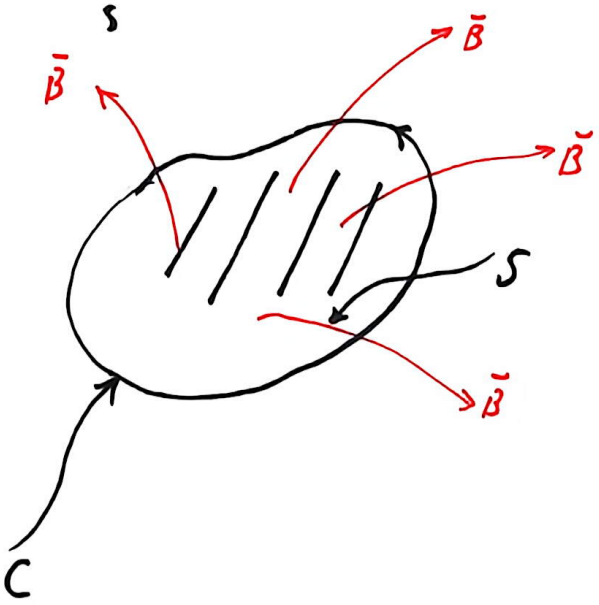
Berry 相位

(三维) $\begin{cases} \vec{A}_n(\vec{R}) \equiv i\langle n,\vec{R}|\nabla_{\vec{R}}|n,\vec{R}\rangle & \text{"矢势"} \\ \vec{B}_n(\vec{R}) \equiv \nabla_{\vec{R}} \times \vec{A}_n(\vec{R}) & \text{"磁场强度"} \end{cases}$

$$\Rightarrow \quad \gamma_n(C) = \oint_C \vec{A}_n(\vec{R}) \cdot d\vec{R} \qquad \text{(Stokes 定理)}$$

$$= \iint_S \left[\nabla_{\vec{R}} \times \vec{A}_n(\vec{R})\right] \cdot d\vec{S}$$

$$= \iint_S \vec{B}_n(\vec{R}) \cdot d\vec{S} \qquad \text{"磁通量"}$$



- $\vec{A}_n^*(\vec{R}) = \vec{A}_n(\vec{R})$; $\vec{B}_n^*(\vec{R}) = \vec{B}_n(\vec{R})$;

$\gamma_n^*(\vec{R}) = \gamma_n(\vec{R})$ (都是实的)

· Berry 相位的规范对称性.

$$\tilde{A}_n(\tilde{R}) = i\langle n, \tilde{R}(t)| \nabla_R | n, \tilde{R}(t)\rangle .$$

· 局域规范变换 (U(1) 规范变换):

$$|n, \tilde{R}(t)\rangle \longrightarrow e^{i\eta(\tilde{R})} |n, \tilde{R}(t)\rangle$$

则, $$\nabla_R |n, \tilde{R}(t)\rangle \longrightarrow \nabla_R \left(e^{i\eta(\tilde{R})} |n, \tilde{R}(t)\rangle \right)$$

$$= i\nabla_R \eta(\tilde{R})\, e^{i\eta(\tilde{R})} |n, \tilde{R}(t)\rangle + e^{i\eta(\tilde{R})} \nabla_R |n, \tilde{R}(t)\rangle .$$

而 $$i\langle n, \tilde{R}(t)| \nabla_R |n, \tilde{R}(t)\rangle \longrightarrow$$

$$i\, e^{-i\eta(\tilde{R})} \langle n\, \tilde{R}(t)| \left[i\nabla_R \eta(\tilde{R})\, e^{i\eta(\tilde{R})} |n, \tilde{R}(t)\rangle + e^{i\eta(\tilde{R})} \nabla_R |n, \tilde{R}(t)\rangle \right]$$

$$= -\nabla_R \eta(\tilde{R}) + i\langle n\, \tilde{R}(t)| \nabla_R |n, \tilde{R}(t)\rangle$$

即 $$\tilde{A}_n(\tilde{R}) \longrightarrow \tilde{A}_n(\tilde{R}) - \nabla_R \eta(\tilde{R})$$

- $\tilde{B}_n(\tilde{R}) = \nabla_{\tilde{R}} \times \tilde{A}_n(\tilde{R}) \longrightarrow$

$$\nabla_R \times \left[\tilde{A}_n(\tilde{R}) - \nabla_R \eta(\tilde{R}) \right]$$

$$= \nabla_R \times \tilde{A}_n(\tilde{R}) = \tilde{B}_n(\tilde{R}).$$

即 $\tilde{B}_n(\tilde{R}) \rightarrow \tilde{B}_n(\tilde{R})$

- $\gamma_n(C) = \iint_S \tilde{B}_n(\tilde{R}) \cdot d\tilde{S} \rightarrow \iint_S \tilde{B}_n(\tilde{R}) \cdot d\tilde{S}$

即 $\gamma_n(C) \rightarrow \gamma_n(C)$

- $\gamma_n(C)$ 与本征态的相位选择无关.

- $\gamma_n(C)$ 为规范不变量. 可导致观测效应.

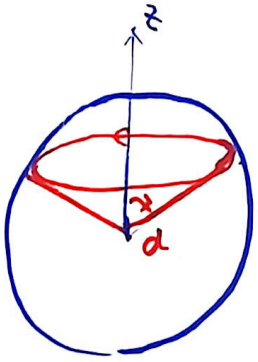
- 参数空间中闭合路径 C 的几何特性决定 $\gamma_n(C)$ 的值，不同闭合路径，只要所得到的"磁通量" $\iint_S \vec{B}_n(\vec{R})\cdot d\vec{S}$ 相同，Berry 相位便相同.

- 例题中，$\gamma_+(t) = (\cos\alpha - 1)\frac{\omega t}{2}$，

  Berry 相位：$\gamma_+(T) = \gamma_+(2\pi/\omega) = \pi(\cos\alpha - 1)$.

  在例题中，改变 B_0 和 ω 的大小，给出相同的 Berry 相位.

- $\gamma_+(C)$ 仅与 $\vec{B}(t)$ 的矢端扫过的回路的几何有关.



立体角 $\Omega = \int \sin\theta\, d\theta\, d\varphi$

$= 2\pi(-\cos\theta)\Big|_0^\alpha$

$= 2\pi(1-\cos\alpha)$

$$\gamma_+(C) = -\frac{\Omega}{2}$$

* 几何相位！

- Berry 理论的实验验证：

  文献：① Phys. Rev. Lett, 1987, 59(3): 251-259

  ② phys. Rev. Lett, 1988, 61: 2030-2033

"Berry phases in Electronic Structure Theory"
David Vanderbilt, Cambridge,
(2018)

$\vec{A}_n(\vec{k})$: Berry connection

$\vec{B}_n(\vec{k})$: Berry curvature

※ $\gamma_n(C)$ 用 $\hat{H}$ 表示

- $\vec{A}_n(\vec{R}) = i \langle n,\vec{R} | \nabla_R | n,\vec{R} \rangle$
- $\vec{B}_n(\vec{R}) = \nabla_R \times \vec{A}_n(\vec{R})$

$$\vec{B}_n(\vec{R}) = i \sum_{m \neq n} \frac{\langle n\,\vec{R} | (\nabla_R \hat{H}) | m\,\vec{R} \rangle \times \langle m\,\vec{R} | (\nabla_R \hat{H}) | n\,\vec{R} \rangle}{[E_n(\vec{R}) - E_m(\vec{R})]^2}$$

$$\gamma_n(C) = \iint_S \vec{B}_n(\vec{R}) \cdot d\vec{S}$$

关系式

$$\vec{B}_n(\vec{R}) = i \sum_{m \neq n} \frac{\langle n\vec{R} | \nabla_R \hat{H}(\vec{R}) | m\vec{R} \rangle \times \langle m\vec{R} | \nabla_R \hat{H}(\vec{R}) | n\vec{R} \rangle}{(E_n(\vec{R}) - E_m(\vec{R}))^2}$$

的推导。

$$\begin{cases} \vec{A}_n(\vec{R}) \equiv i \langle n\vec{R} | \nabla_R | n\vec{R} \rangle . \\ \vec{B}_n(\vec{R}) \equiv \nabla \times \vec{A}_n(\vec{R}) . \end{cases}$$

(以下推导中，略去所有 $\vec{R}$)

$$\begin{aligned} \vec{B}_n = \nabla \times \vec{A}_n &= i \nabla \times \left[\langle n | \nabla | n \rangle \right] \\ &= i \left[(\nabla \langle n |) \times \nabla | n \rangle \right] \\ &\quad + i \left[\langle n | \nabla \times \nabla | n \rangle \right] \\ &\quad \quad \nabla \times \nabla = 0 \\ &= i \left(\nabla \langle n | \times \nabla | n \rangle \right) . \\ &= + i \sum_m \left((\nabla \langle n |) | m \rangle \times \langle m | \nabla | n \rangle \right) \quad \cdots (A) \end{aligned}$$

• 将上式用 $\hat{H}$ 表示.

$$\hat{H}|n\rangle = E_n|n\rangle ,$$

$$\Rightarrow \quad \nabla\left[\hat{H}|n\rangle\right] = \nabla\left[E_n|n\rangle\right],$$

$$\Rightarrow \quad (\nabla\hat{H})|n\rangle + \hat{H}(\nabla|n\rangle) = (\nabla E_n)|n\rangle + E_n\nabla|n\rangle ,$$

• 上式两边左乘 $\langle m|$, 得 $(m\neq n)$

$$\langle m|(\nabla\hat{H})|n\rangle + \langle m|\hat{H}\nabla|n\rangle = E_n\langle m|\nabla|n\rangle .$$

$$\Rightarrow \quad \langle m|(\nabla\hat{H})|n\rangle + E_m\langle m|\nabla|n\rangle = E_n\langle m|\nabla|n\rangle .$$

$$\Rightarrow \quad \langle m|(\nabla\hat{H})|n\rangle = (E_n - E_m)\langle m|\nabla|n\rangle$$

$$\Rightarrow \quad \langle m|\nabla|n\rangle = \frac{\langle m|(\nabla\hat{H})|n\rangle}{E_n - E_m} \qquad \cdots(B)$$

交换 n 与 m, 得

$$\langle n|\nabla|m\rangle = \frac{\langle n|(\nabla\hat{H})|m\rangle}{E_m - E_n} \qquad \cdots(C)$$

由 $\langle n|m\rangle = \delta_{mn}$ 得

$$(\nabla\langle n|)|m\rangle + \langle n|\nabla|m\rangle = 0.$$

$$\Rightarrow \quad (\nabla\langle n|)|m\rangle = -\langle n|\nabla|m\rangle .$$

$$\Rightarrow \quad -(\nabla\langle n|)|m\rangle = \frac{\langle n|(\nabla\hat{H})|m}{E_m - E_n} \qquad \cdots(D)$$

将 (B),(D) 式代入 (A) 式中. 得

$$\tilde{B}_n = i\left((\nabla\langle n|)|n\rangle \times \langle n|\nabla|n\rangle\right)$$

$$+ i\sum_{m\neq n}\left\{\frac{\langle n|(\nabla\hat{H})|m\rangle\times\langle m|(\nabla\hat{H})|n\rangle}{(E_n - E_m)^2}\right\}$$

$$= i\left(-\langle n|\nabla|n\rangle \times \underbrace{\langle n|\nabla|n\rangle}_{=0}\right)$$

$$+ i\sum_{m\neq n}\frac{\langle n|(\nabla\hat{H})|m\rangle\times\langle m|(\nabla\hat{H})|n\rangle}{(E_n - E_m)^2} \quad \Rightarrow$$

$$\tilde{B}_n(\tilde{R}) = i \sum_{m \neq n} \frac{\langle n,\tilde{R}| (\nabla_R \hat{H}(\tilde{R})) |m,\tilde{R}\rangle \langle m,\tilde{R}| (\nabla_R \hat{H}(\tilde{R})) |n,\tilde{R}\rangle}{(E_n(\tilde{R}) - E_m(\tilde{R}))^2} \quad \cdots (E)$$

(E) 式中. $\hat{H}(\tilde{R}(t)) = \hat{H}(\tilde{B}(t)) = \frac{e}{m} \tilde{B}(t) \cdot \tilde{S}$

$\Rightarrow$ $\nabla_B \hat{H}(\tilde{B}) = \frac{e}{m} \tilde{S}$